

KIRUTHIKA S

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CAREER OBJECTIVE

Final-year Electronics and Communication Engineering student with hands-on exposure to PCB design, schematic capture, embedded hardware and basic circuit testing. Familiar with KiCad, component selection, footprint assignment, PCB layout and routing, along with Arduino and STM32-based hardware development. Seeking an entry-level Electronics/Hardware Trainee Engineer role to apply my electronics fundamentals and develop practical engineering skills.

EDUCATION

Qualification	Institution	Year	Score
B.E. Electronics and Communication Engineering	Kongunadu College of Engineering and Technology	2023-Present	CGPA 9.34
HSC	Sowdambikaa Hr. Sec. School, Thottiyam	2023	86.16%
SSLC	Sowdambikaa Hr. Sec. School, Thottiyam	2021	All Pass

INTERNSHIP

1. **Embedded Systems - CodeBind Technologies, Trichy** **15 days**

   Hands-on experience with sensors, actuators, and Arduino through a "Disaster Alerting System" project.

1. **Embedded Systems & IoT - Taras Systems and Solutions, Bengaluru** **6 Months**

   Built a "Smart Priority-Based Load Control System" using STM32 and MSP430.

2. **VLSI - Univision, Chennai** **1 Month**

   Learned about Verilog HDL and System Verilog & UVM basics

TECHNICAL SKILLS

- Digital Electronics, Analog Electronics, Basic Circuit Design
- **Programming Languages:** C
- **PCB Design:** KiCad, Schematic Capture, PCB Layout & Routing, Component Selection & Placement, Footprint Assignment, THT/SMD Components, DRC
- **Embedded Hardware:** Arduino, STM32, Sensor Interfacing
- **Communication Protocols:** UART, SPI, I2C
- **FPGA:** FPGA Board, Vivado
- **HDL:** Verilog, SystemVerilog
- **Tools & Platforms:** KiCad, LTSpice, Arduino IDE and STM32CubeIDE

PROJECTS

Project 1

Title: Smart Medicine Reminder and Expiry Detection System.

Tools Used: Arduino Uno, VS Code, MediaPipe, PySerial, Arduino IDE, OpenCV.

Description: The Smart Medicine Reminder and Expiry Detection System helps users take medicines on Time through alerts and it also predicts expiry dates to ensure safety.

Project 2

Title: Smart Priority-based Load Control System Using STM32.

Tools Used: STM32CubeIDE, STM32WB55 Microcontroller, ACS712 Current Sensor, Relay Module, Load (LED / Fan) and Buzzer.

Description: This project monitors the electrical load and manage power distribution intelligently. When an overload occurs, it automatically disconnects low-priority loads while keeping important loads active, ensuring safe and efficient power management.

CERTIFICATIONS

- ✓ PLC – Value Added Course
- ✓ PCB – Value Added Course
- ✓ NPTEL Certification – Introduction to Industry 4.0 and Industrial Internet of Things (IIoT)
- ✓ Certificate for Completion of C Training – Spoken Tutorial Project, IIT Bombay
- ✓ Secured First Class with Distinction in English Junior & Senior Typewriting.

HACKATHONS, ACHIEVEMENTS & PRESENTATIONS

- Won **First Prize and a cash award** in a Biomedical Department Hackathon for a Blockchain-based project at KNCET.
- Won **First Prize with a cash award** in an ECE Department Hackathon for a "Rotten Egg Detection System" project at KNCET.
- Won **First Prize with a Cash Award** in Idea Presentation organized by IETE.
- Secured **Second Prize with Cash Award** in Idea Presentation at Kumarasamy College.
- Participated in the **Hackathons** like SIH, MSME and Hack at Train conducted by KNCET.

AREA OF INTERESTS

- ❖ Electronics Hardware
- ❖ PCB Design
- ❖ Embedded Systems
- ❖ Electronics Testing and Troubleshooting

HOBBIES

- ➢ Singing

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge.

SIGNATURE

KIRUTHIKA S